Shift-Robust Candidate Cell Sets for Handover via Adaptive Conformal Prediction

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***Abstract*—Predictive handover models can look accurate on source data and still fail once mobility, horizon, or neighbor measurements shift. We study conformal prediction for *candidate cell sets* under shift, not just top-1 prediction. The benchmark has two parts: synthetic A3/TTT replay for controller realism, and repeated real-trace evaluation on Irish 5G driving logs with speed-split, random trace-holdout, and chronological holdout. On synthetic regime shift, static CP reaches 73.8% coverage while DACI reaches 91.8%, but HO success only moves from 56.0% to 56.4%. This is the first main finding: better set reliability does not automatically mean large downstream controller gains. On Irish repeated splits, DACI consistently beats static CP by $+14.7$ coverage points on speed-split (95% CI $[13.3, 16.1]$), $+3.6$ on trace-holdout ($[-2.4, 9.6]$), and $+18.3$ on chronological holdout ($[14.5, 22.2]$). Worst confidence-decile coverage also improves from 47.2% to 81.4% on speed-split and from 47.9% to 60.4% on chronological holdout. RAPS is the strongest non-adaptive baseline. A fixed budgeted DACI setting becomes a real lower-load alternative: it matches DACI on speed-split at lower load, improves trace-holdout, and loses only 0.7 points on chronological holdout while still reducing load. The main result is simple: adaptivity matters most in the hard real slices, and handover papers should report reliability together with measurement cost.**

I. Introduction

Predictive handover is attractive because it can reduce interruption and avoid late switching. The hard part is not point accuracy on source data. The hard part is staying reliable after the environment changes. Speed changes, stale neighbor reports, and horizon mismatch all change the link between current features and the future best cell. Traditional A3-style rules handle part of this with hysteresis and TTT [2], [3], but they do not give explicit reliability control. Standard classifiers improve accuracy but do not tell the controller when predictions are no longer safe.

Conformal prediction (CP) is a natural fit because it returns a set of plausible next cells instead of one label, with explicit coverage control [5], [6]. Recent wireless CP work studies calibration and beam prediction [7]–[9], but the evidence base for handover is still thin. In particular, three gaps remain.

First, many studies stop at set coverage and do not test how a controller behaves under A3/TTT replay. Second, adaptive methods are rarely tested with delayed feedback or small recalibration budgets. Third, real-trace studies often report average coverage only, not the worst slices where deployment failures appear.

This paper closes those gaps with one narrower claim than the earlier draft. The paper is mainly about *shift-robust candidate-cell reliability*. Synthetic replay explains the mechanism. Irish traces show when adaptivity helps on real drift. The main contributions are:

1) A `v7` benchmark with synthetic A3/TTT replay, delayed feedback, recalibration stress, and repeated Irish split evaluation.
2) A clear empirical result: DACI gives stable real-data reliability gains under speed, random, and chronological drift, while controller KPIs move much less than set coverage in synthetic replay.
3) A worst-slice analysis showing where static CP fails, where RAPS stays competitive, and where adaptivity matters most.
4) A direct budget check with fixed-K and truncated-static baselines, showing that simple truncation does not explain the adaptive gains.

II. Related Work

Most handover prediction work optimizes top-1 decisions or switching rules, for example via reinforcement learning or conditional handover logic [4], [12]. That literature usually reports accuracy, interruption, or switching behavior, not calibrated candidate-cell reliability under shift.

Wireless conformal work is stronger on calibration than on handover. Recent papers study uncertainty calibration for wireless models and reliable beam selection [7]–[9], while adaptive conformal inference under shift is established in the broader ML literature [10]. Our paper sits at that intersection: handover, real drift, and adaptive conformal candidate sets with explicit cost reporting.

III. Problem Setup and Methods

At time t, the model predicts the best future cell after horizon H:

$$y_t = \arg\max_k \mathrm{RSRP}_k(t + H). \tag{1}$$

For the static and adaptive CP methods we use the nonconformity score

$$s_t = 1 - \hat{p}(y_t \mid x_t), \tag{2}$$

calibrate a quantile q_t, and return the candidate set

$$\mathcal{C}(x_t) = \{k : 1 - \hat{p}(k \mid x_t) \le q_t\}. \tag{3}$$

Coverage measures whether the future best cell falls inside the set. Set size maps to measurement cost. RAPS uses its

regularized cumulative score instead of Eq. (3), but it is evaluated with the same coverage and cost metrics.

We keep the main adaptive and non-adaptive methods, plus two simple cost baselines used only to test whether budget gains come from trivial truncation.

A3/TTT baseline. Synthetic replay uses a full-scan A3 controller with 3 dB offset, TTT of 2 steps, and minimum dwell of 2 steps. Irish traces do not log full neighbor measurement reports, so the real-trace baseline is a serving-cell retention proxy.

Static CP. One source-calibrated threshold is frozen for the full target stream.

RAPS CP. RAPS regularizes large sets while still calibrating on source data.

Triggered ACI. Static CP is used on confident samples; ACI is used below a source-calibrated confidence trigger.

DACI. Let $e_t = \mathbb{1}\{y_t \notin \mathcal{C}(x_t)\}$ and let $\bar{e}_t$ be an EMA of recent errors. DACI uses

$$\alpha_{t+1} = \text{clip}\left(\alpha_t + \gamma_t(\alpha - e_t)\right), \tag{4}$$

with $\gamma_t = \gamma_{high}$ when $\bar{e}_t > \alpha$ and $\gamma_t = \gamma_{low}$ otherwise [10].

Budgeted DACI. Budgeted DACI adds a size-ratio term $r_t = |\mathcal{C}(x_t)|/K$ and updates

$$\alpha_{t+1} = \text{clip}\left(\alpha_t + \gamma_t(\alpha - e_t) + \lambda(r_t - \rho)\right), \tag{5}$$

where ρ is the target size ratio and λ is the budget gain.

Fixed-K and static-CP truncation. Fixed-K always measures the top K_{max} cells. Static-CP truncation caps static CP at the same K_{max}. These are not headline methods. They are sanity checks for the budget story.

IV. V7 Benchmark Design

A. Synthetic Replay Benchmark

The source environment uses a 4×4 cell grid, shadowing standard deviation 6 dB, measurement noise 4 dB, speed range 1–30 m/s, and prediction horizon $H = 10$. We train on source trajectories, calibrate on held-out source trajectories, and evaluate only on target trajectories. The final `v7` run uses 3 seeds (42, 123, 456), 400 trajectories per seed, 15 training epochs, and replay cap $K_{max} = 12$.

We evaluate five target drifts:

1) speed shift: 20–50 m/s;
2) shadow shift: shadowing standard deviation 10 dB;
3) horizon shift: target horizon $H = 20$;
4) stale-neighbor shift: 50% of non-serving neighbor measurements corrupted by 8 dB staleness noise;
5) regime switch: source-like first half and harsh second half.

The replay controller reports coverage, HO success, measurement load, handover rate, ping-pong rate, wrong-cell dwell, and interruption proxy. The main figures are generated directly from `figures/shift-results-v7.json`.

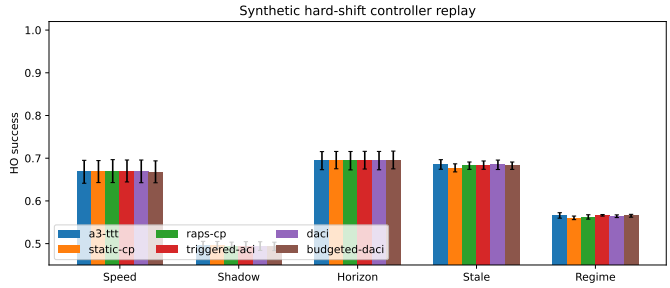

Fig. 1. Synthetic controller replay on hard shifts. Adaptive methods raise set coverage sharply under shift, but downstream HO success changes much less.

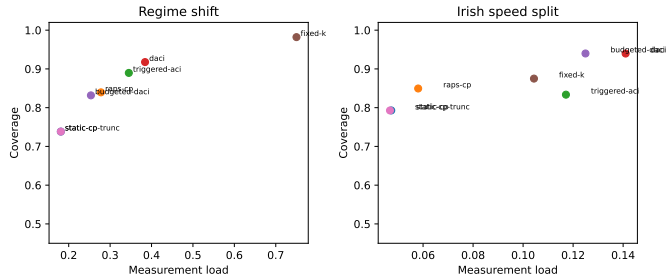

Fig. 2. Reliability-cost frontiers for synthetic regime shift and Irish speed-split drift. The useful question is not only which method is most reliable, but how much measurement cost buys that reliability.

B. Irish 5G Real-Trace Benchmark

We use the Irish 5G driving dataset [13]. The final `v7` run uses 40 driving traces with horizon $H = 10$ and repeated split seeds $\{42, 123, 456, 789, 2024\}$. We keep three protocols.

Speed-split. The lower-speed 20 traces are the source pool and the higher-speed 20 traces are the target pool.

Trace-holdout. We randomly split 32 source traces and 8 target traces.

Chronological holdout. The earliest 32 traces form the source pool and the latest 8 traces form the target pool.

The Irish logs do not include full per-step neighbor measurement matrices needed for faithful A3/TTT replay. For that reason, the real-trace study focuses on future-cell coverage, set size, measurement load, delayed-feedback robustness, recalibration budgets, and worst-slice reliability. This split is deliberate: synthetic data answers controller realism, Irish data answers deployment drift.

V. Results

A. Hard-Shift Controller Replay

Table I is the mechanism check, not the headline claim. Static CP is not enough once shift becomes hard. On shadow shift, coverage rises from 69.1% to 90.4% when moving from static CP to DACI. On regime switch, coverage rises from 73.8% to 91.8%.

The important point is what *does not* move much. On regime switch, HO success only moves from 56.0% for static CP to 56.4% for DACI. This means candidate-set reliability and

TABLE I
SYNTHETIC HARD-SHIFT RESULTS

Shift	Method	Coverage (95% CI)	HO Success (95% CI)	Load (95% CI)
Shadow	Static CP	.691 [.676, .707]	.497 [.485, .508]	.164 [.153, .175]
	RAPS CP	.799 [.788, .810]	.494 [.481, .507]	.249 [.248, .250]
	Triggered ACI	.864 [.861, .868]	.494 [.480, .509]	.384 [.368, .400]
	DACI	**.904** [**.899**, **.910**]	.494 [.480, .508]	.405 [.381, .428]
	Budgeted DACI	.817 [.813, .822]	.494 [.479, .508]	.268 [.256, .280]
Regime	Static CP	.738 [.713, .764]	.560 [.555, .566]	.181 [.169, .193]
	RAPS CP	.840 [.836, .843]	.562 [.555, .570]	.278 [.274, .282]
	Triggered ACI	.890 [.887, .893]	**.566** [**.564**, **.569**]	.345 [.336, .354]
	DACI	**.918** [**.909**, **.927**]	.564 [.560, .569]	.384 [.368, .401]
	Budgeted DACI	.832 [.828, .835]	.565 [.557, .573]	.254 [.251, .258]

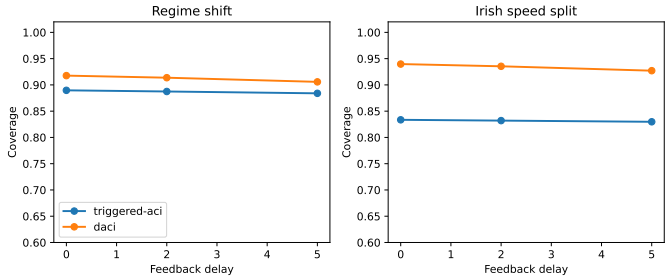

Fig. 3. Coverage under delayed feedback. DACI degrades slowly as label delay grows, which makes it a useful adaptive baseline.

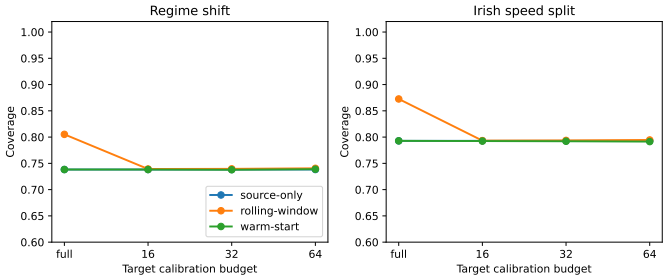

Fig. 4. Coverage under recalibration budgets. Full rolling updates help on synthetic regime switch, while tiny warm-start budgets do little. Irish gains are smaller because the source split is already fairly strong.

TABLE II
IRISH REPEATED-SPLIT RESULTS

Method	Speed	Trace-Holdout	Chronological
	Coverage / Load	Coverage / Load	Coverage / Load
Static CP	.793/.047	.906/.078	.691/.043
RAPS CP	.849/.058	.924/.140	.763/.057
DACI	**.940**/**.141**	.942/.097	**.874**/**.171**
Budgeted DACI	.940/.125	**.961**/**.095**	.867/.159

downstream controller gain are related, but not interchangeable. That is why the rest of the paper centers on real-data reliability and cost, not on a stronger handover-control claim than the evidence supports.

The direct budget checks are also useful. Fixed-K reaches 98.2% coverage on regime shift, but only by paying measurement load 0.75. Static-CP truncation is almost unchanged from static CP. Budgeted DACI reduces load from 0.384 to 0.254 versus DACI, but also drops coverage from 91.8% to 83.2%. The budget idea is promising, but DACI remains the main method.

B. Delayed Feedback and Recalibration

Delayed feedback matters because online labels do not arrive instantly in real systems. Figure 3 shows that DACI stays stable: on regime shift it moves from 91.8% coverage at delay 0 to 90.6% at delay 5. This is a support result, not the headline. It says the adaptive method is not fragile once feedback slows down.

Recalibration is more subtle. On synthetic regime shift, source-only CP stays at 73.8% coverage, while full rolling recalibration reaches 80.5%. Small target budgets of 16, 32, or 64 points are not enough to close the gap in this hard nonstationary setting. On Irish drift, the source split is already fairly strong, so recalibration helps less.

C. Irish Real-Trace Drift

The Irish results are the main result of the paper. Table II and Figure 2 show three clear patterns.

First, DACI is the only method with stable gains across all three real-data protocols. On speed-split drift, DACI improves coverage over static CP by 14.7 points (95% CI $[13.3, 16.1]$). On trace-holdout, the gain is 3.6 points ($[-2.4, 9.6]$). On chronological holdout, which is the hardest and most realistic split in this paper, the gain is 18.3 points ($[14.5, 22.2]$). A simple example is the chronological split: static CP reaches 69.1% coverage, DACI reaches 87.4%.

Second, RAPS is the strongest non-adaptive baseline. It consistently improves on static CP at very low extra load, but it stays below DACI on all three real-data protocols.

Third, budgeted DACI is now a credible secondary result, but not the headline. With a fixed 10% target budget ratio and $\gamma = 0.01$, it matches DACI on speed-split while reducing average set size by 1.8 (95% CI $[-2.4, -1.3]$). On trace-holdout it improves coverage by 1.8 points with slightly lower average set size, although the coverage interval still overlaps zero. On chronological holdout it loses 0.7 points versus DACI (95% CI $[-2.3, 0.8]$) while reducing average set size by 1.3. The fixed-K baseline still matters here: it can buy more coverage with more load, so DACI remains the main method.

D. Worst-Slice Reliability

Average coverage hides the cases reviewers care about most. The worst confidence decile is where static CP breaks first. On speed-split, worst-decile coverage rises from 47.2% for static CP to 81.4% for DACI, with repeated-split intervals $[32.2, 62.2]$ and $[75.7, 87.2]$. On trace-holdout, it rises from 68.5% to 82.9%. On chronological holdout, it rises from 47.9% to 60.4%. The intervals are still wider than the mean-coverage intervals, but the pattern is consistent: adaptivity helps most in the slices where deployment failures are most likely.

Static-CP truncation is also informative here. It is almost identical to static CP across the Irish protocols. So the gain is not coming from a trivial cap on set size. It is coming from adaptation.

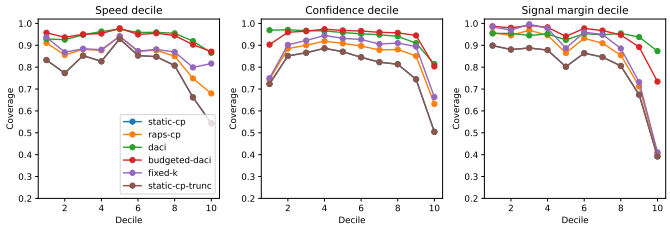

Fig. A1. Irish worst-slice diagnostics. The lowest-confidence and lowest-margin deciles are where static CP fails first and adaptive methods matter most.

VI. Reproducibility

The canonical artifact file is `shift-results-v7.json`. The main figures are `controller-frontier-v7.pdf`, `delayed-feedback-v7.pdf`, `recalibration-budget-v7.pdf`, and `irish-worst-slice-v7.pdf`. The one-command rebuild is `uv run python run_paper_v7.py`. It reruns the pipeline, validates the schema, runs the controller tests, and rebuilds the PDF.

VII. Discussion and Conclusion

This paper is strongest as a candidate-set reliability paper, not as a large handover-control paper. The synthetic replay is useful because it shows why: coverage can move a lot while HO success moves only a little.

The real-data result is much cleaner. DACI consistently improves reliability under speed, random, and chronological drift. RAPS is the strongest non-adaptive baseline. Budgeted DACI now gives a cleaner lower-load alternative, but DACI still has the strongest hardest-split result and the best worst-slice robustness. Simple truncation does not explain the adaptive gains.

The main lesson is simple: handover papers should separate three questions. Can the set stay reliable under shift? Can the controller stay stable? How much measurement cost buys that reliability? Once those questions are separated, the value of adaptive conformal prediction becomes much clearer.

Limitations. The real-data evidence still comes from one Irish dataset, with repeated splits rather than a second independent corpus. The logs also do not expose the full neighbor-state history needed for faithful A3/TTT replay. Finally, the budgeted controller is only tested with a fixed global setting, not protocol-specific tuning.

The next step is not another small CP variant. It is richer downstream evaluation with either more real traces or a stronger simulator such as 5G-LENA or Simu5G.

Appendix A
Worst-Slice Diagnostics

Figure A1 is the main appendix diagnostic. It shows why the speed-split result is not just a high average from easy samples. Static CP loses reliability badly in the hardest confidence slice. DACI and RAPS stay much flatter. Budgeted DACI helps on some hard slices, but not enough to replace DACI in the main story.

TABLE A1
Final V7 Defaults

Setting	Value
Synthetic seeds	42, 123, 456
Synthetic trajectories per seed	400
Synthetic training epochs	15
Target miscoverage α	0.10
Triggered-ACI quantile	0.7
DACI $(\gamma_{low}, \gamma_{high}, \beta)$	$(0.005, 0.02, 0.95)$
Replay A3 offset	3 dB
Replay TTT	2 steps
Replay minimum dwell	2 steps
Replay cap K_{max}	12 cells
Irish horizon	10 steps
Irish max traces	40
Irish split seeds	42, 123, 456, 789, 2024
Budgeted DACI target ratio	0.10
Budgeted DACI γ	0.01

Appendix B
Settings and Assumptions

The main assumption in the paper is that synthetic replay is the right place to test controller realism, while the Irish traces are the right place to test real drift and slice-wise reliability. The main risk is that Irish logs do not expose the full measurement state needed for a fully faithful A3/TTT replay. That is why the paper does not over-claim the Irish controller metrics.

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